



KANDIDAT

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PRØVE

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Informasjonssystemer

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Tasks

Oppgave	Tittel	Oppgavetype
i	Exams Cover	Informasjon eller ressurser
i	The Amazon case	Informasjon eller ressurser
1	Task 1 (25 points)	Langsvar
2	Task 2 (20 points)	Langsvar
3	Task 3 (10 points)	Langsvar
4	Task 4 (5 points)	Langsvar
5	Task 5 (15 points)	Langsvar
6	Task 6 (5 points)	Langsvar

¹ Task 1 (25 points)

1.1 (5 points)

In a BPMN model, pools represent the major participants (often distinct organizations or systems) in the process, while lanes represent roles, departments, or specific functions within an organization or system. Together, pools and lanes help delineate distinct roles and responsibilities within the overall process.

In the BPMN model of the online shopping and delivery process, the names of the pools and lanes are missing. Assign appropriate names to them (based on the description of the case) to ensure clarity in the workflow and help identify points of interaction between participants. Indicate where in the model are those pools and lanes located.

Keep your answer between 0-150 words.

1.2 (20 points)

The BPMN model contains several mistakes, both syntactic and semantic. Find at least two syntactic and two semantic mistakes (so, four mistakes in total). Describe where in the model those mistakes are located. Provide an explanation for why you consider each to be a mistake.

Note: the missing names of pools and lanes from 1.1. do not count as mistakes.

Keep your answer between 0-250 words.

Fill in your answer here

1.1

- Top lane: *Credit card company*
- Middle lane: *Customer*
- Bottom left lane: *Fulfillment department*, whereas the top lane are named *picker* and the bottom lane are named *packager*
- Bottom right lane: *Carrier*

1.2

Syntactic:

- **AND-gateway in picker lane:** only has one outgoing sequence flow. It is supposed to serve either as a concurrent thread or as a synchronizing join.
- **Send to carrier truck:** is a manual task that has a connection with another manual task through a message flow. This is not allowed. The message flow should have been sent to an intermediate task like it has been done previously throughout the model.
- **Gateway in carrier lane:** the gateway has a circle within it, which is not a valid symbol according to BPMN modeling. Valid symbols are nothing (clear), + and x.

Semantic:

- **The connection between picker and packager lane:** it is not clear that the bin is passed to the Amazon packager. Though, this is mostly due to the syntactic error described earlier.
- **The intermediate task in the docker:** it connects send to carrier truck from the packager lane and is a "send" intermediate task. This should have been a receive, as it is stated that the carrier collects the package.
- **Deliver items activity:** the connection between the carrier pool and the customer pool should have been done with a *timer* intermediate event. The way it is modeled now it appears the customer receives the package right away, which is not correct.

Ord: 243

2 Task 2 (20 points)

As explained, the human roles play a critical part in ensuring the smooth processing and fulfillment of the customer's order.

2.1 (15 points)

What types of Business Information Systems do you think are needed to support the task of each of the following participants?

- (a) Credit card company;
- (b) Amazon fulfillment department;
- (c) Carrier dock

Name two types of IS per type of participant, so **for each participant you get 2 points**. Elaborate on your answers. **For each participant you get 3 points**.

Keep your answer between 0-250 words.

2.2 (5 points)

A Fraud Detection System (FDS), which is a type of DSS, uses data analysis to detect and flag unusual transaction patterns that might indicate fraudulent activities, helping to secure transactions and minimize risk. Which of the above participants needs an FDS – in addition to the two other types you identified in 2.1 (**2 points**)? Explain your answer (**3 points**). **Note:** you cannot use the FDS as answer to 2.1.

Keep your answer between 0-150 words.

Fill in your answer here

2.1 (a)

- A Transaction processing system (TPS) is crucial due to the need to record completed business transactions. Without one, this would not be possible, given that a credit card company handles an enormous amount of business transactions daily. This data is also very structured, as each payment requires the same details.
- Customer Relationship Management (CRM) system could also prove to be valuable. For the credit card company, there are many competitors. So to keep the customer using their service, they should employ a CRM to track and manage all customer encounters to increase customer satisfaction. Here, for this specific case, failed transactions should be tracked and serve as a KPI where the goal is zero failed transaction due (eg. due to system malfunctions).

2.1 (b)

- Management information system (MIS) would be deemed as needed. This is because it focuses on operational efficiency, which is valuable for the fulfillment department. A manager could receive a report from the MIS with details regarding inventory needed (demand reports), "svinn" (exception reports) and improving processes (developing effective reports). In other words, this system helps to do things right. Here, whereas staffing is highly important, it would be very useful for management decision making. It could also be used for paychecks, employee work hours and holiday scheduling. It should be noted that the data is structured, which supports the choice for a MIS.
- Supply-Chain Management system (SCM) could be very valuable. The department should know what they will package to optimize the schedule (eg. get all resources ready). The value-chain itself includes multiple aspects like inbound logistics (eg. orders), warehouse and storage, production, finished product storage, outbound logistics (eg. sent out orders). These serve as pinpoints for further improvement of the process within the fulfillment department. Perhaps also included the carrier if multiple carrier options are present.
- * A TPS could also be employed here. It can support order processing, which is needed throughout the fulfillingment departments job. This is because they execute the packaging and labeling needed for shipping.

2.1 (c)

- A Decision support system (DSS) helps to provide rapid information, handle large amounts of data and provide both textual and graphical orientation. In this scenario I assume that a carrier transports more than one package at the time. Given that there are many packages, it is hard to determine which ones to deliver first, and what route to drive. Therefore, a DSS could help solve this complex and unique problem, suggesting alternatives and assisting in final decision making. It should also be noted that the data is semistructured, since you could order from Amazon in different countries (different zip-codes and logic) because in this model it is not clear how they sort the bins and if the carrier receives solely packages in the same area or multiple.
- Knowledge information system (KMS) is from my experience in the carrier industry highly needed. As a new employee within the business, there are a lot to learn, whereas a knowledge management system is in place to serve this need. Here, the system share and process knowledge, and employees could learn new knowledge. Through that procedures and approaches may be changed. It could be as simple a SharePoint platform.
- * A CRM should definitely be employed. However, this is not directly supported by the outlined model, since it does not state that the customer may file complaint regarding the packaging (eg. if the package got damaged during transportation). This could have been a good addition to make sure that all complaints are followed up, and resolved.

* An Enterprise resource planning (ERP) system could be determined as valuable given the need to support manufacturing and finance. Customer demand could be forecasted to make sure all needed packaging resources, enough people and may also determine what needs to be ordered to meet the schedule. Personally, I excluded ERP due to it being extremely large - and not fitting under one specific participant.

2.2

The Credit card company would be the one needing a Fraud Detection System (FDS). Given that the credit card company takes multiple transactions each day, there has to be used some kind of data analysis to detect and flag anomalies. These transactions may sometimes involve a lot of money, and transaction patterns picked up by the FDS may pick up these fraudulent activities. There is also a lot of legal demands around the banking industry stating that they need to have in-place some security measures revolving frauds. Additionally, Traditional TPS systems can support purchasing. Given the possible processing capabilities, some kind of security mechanisms is needed.

Ord: 762

Maks poeng: 20

3 Task 3 (10 points)

3.1 (5 points)

The differentiation between data, information, and knowledge helps each participant make better decisions at every stage of the order, fulfillment, and delivery process in the case described above. What could be data, information, and knowledge in the context of the Amazon shopping and delivery process? Give two examples of data, two examples of information and one example of knowledge for any of the tasks described in the case. **Each of your answers gets 1 point.**

Keep your answer between 0-150 words.

3.2 (5 points)

We talked about the dimensions of valuable information. List 3 that you believe are relevant for the case of credit card transactions in the above case (**1 point for each**) and briefly explain and justify why *two of them* are important (**1 point for each**).

Keep your answer between 0-150 words.

Fill in your answer here

3.1

Data are raw facts. Some examples are the payment data (which by itself are just numbers) and the number cost of the items. These could have been information if they were provided contextual meaning. Information are data that are organized in a way that they provide extended insights rather than the data itself. Some examples could be the order details (that probably includes the measurement details in inches. Note that the numbers itself would be data if it was not followed by inches), the response the customer receives when the payment fails (could be followed by why it was failed, perhaps no service was available) or the scheduled day-hours for a specific employee. Knowledge are something that is built-up through information and interrelations of information with for example experiences. Here, if the payment process fails, the customer has to pay once more. The knowledge here is something beyond the model itself, that the customer probably has to transfer more money to his card, or that he realize that he does not have money on his card.

3.2

There are in total eleven characteristics in the book regarding valuable information. Of these, I regard flexible, complete and accurate relevant for the case of interest. Flexible regards that something can be used for several purposes. In this case the case of credit card transactions may only serve as a minor actor for our model. Complete refers to that all important information is included, but not anything more than is needed (Ocham's razor). This ensures that no redundant information is included. Since the credit card company is not an actor that is directly involved in the Amazon business, there are definitely aspects that occur that we don't see. However, of those related to us, those are modeled, and that is if the payment is accepted or not. In other words, it is complete. Accurate involves that the information is error free. By following false information further errors may occur, like executing wrong packaging procedures. According to the case given, the information is error free. The element of payment is important, in some cases Amazon could not have provided their service if no payment was received or perhaps they don't offer a "pay later service".

Ord: 372

Maks poeng: 10

4 Task 4 (5 points)

Amazon's online shopping and delivery process involves multiple participants, each relying on accurate, real-time data to perform their tasks. Currently, these participants operate with limited visibility into the overall workflow, leading to possible inefficiencies such as delays in order fulfillment, mismatched inventory levels, and inconsistent delivery updates. Which kind of information system (**except from an ERP**) should be used to for real-time monitoring and reporting across its shopping and delivery processes, using available transaction information? Motivate your answer.

Keep your answer between 0-50 words.

Fill in your answer here

4

Management Information System (MIS) is an optimal choice here. This system uses TPS data as input (structured data that is easily programmable). The details may be formatted into the preferred format (eg. monitoring) and it allows for daily checks fulfilling the real-time demand. For Amazon, it is important to state that these system handles big data, which is huge amounts of data that makes it hard to process using traditional database technology.

Ord: 73

Maks poeng: 5

5 Task 5 (15 points)

Amazon emphasizes efficiency, scalability, and customer satisfaction in its online shopping and delivery processes. Therefore, the organization is considering automating warehouses with robots, or experimenting with drone deliveries.

5.1 (10 points)

Those are major organizational changes. We have discussed two ways of doing this: re-engineering and continuous improvement. Which of these two approaches would you consider to be most suitable for the above changes? Elaborate on your answer, also explaining why the other option is less suitable.

Keep your answer between 0-150 words.

5.2 (5 points)

Those are not only major organizational changes, but also innovations. We discussed two types of innovations. Which type of innovation is the warehouse automation and which type of innovation is the drone delivery? Elaborate on your answers.

Keep your answer between 0-150 words.

Fill in your answer here

5.1

Reengineering is a case of a radical redesign to improve for instance the business processes or the organizational structure. It is also called process redesign. In some cases it aims to achieve reduced delivery times (drones) and increase product service and quality (automation with robots, packaging). In this case, swapping out the staffing, as I assume it, with robots or experimenting with drone deliveries is major change indeed, and a reengineering case. The initiative is probably considered by the manager to stay competitive by increasing revenues, profitability and customer satisfaction. Continuous improvement refers to that business processes can add value to products or services. This is done through iterative improvement as the name states. However, from my understanding, I assume that the workers are replaced by the suggested alternatives. This makes me believe it is a top-down change, therefore not thinking it is this idea, but rather reengineering.

5.2

An innovation is either a new product or doing something in a different way than it has been done normally. In this case we have one innovation revolving automation in the warehouses through robots and another one with drone deliveries. Neither of them has been done before, at least from what we are informed. The first one can be labeled as a product innovation, since it is something the company directly employs, and does not directly benefit anyone else than themselves, like financial savings in production. The drone deliveries affect the customer directly given that if the delivery goes faster this way (assumingly). Thus, this is a benefit for this individual, therefore this can be defined as a service innovation. I am also led to believe that both innovations are closed, since Amazon is a private company that would not want others to earn as much on the innovation as they potentially would.

Ord: 303

Maks poeng: 15

6 Task 6 (5 points)

Amazon's online shopping and delivery is a Business-2-Consumer (B2C) e-commerce activity. We discussed three additional different types of e-commerce. List them and for each, give a real-life example.

Keep your answer between 0-50 words.

Fill in your answer here

6

- Business-2-business (B2B) is a subset of e-commerce where all participants are organizations. An example is Business Travel Direct, and a lot of other travel agencies.
- Consumer-2-consumer (C2C) is a subset of e-commerce where all participants are customers. Examples may include Tise, Ebay, Finn.no
- E-government consists of Government-2-consumer (G2C), Government-2-business (G2B) and Government-2-government (G2G). An example of G2C could be Skatteetaten with filing taxes.

Ord: 64

Maks poeng: 5